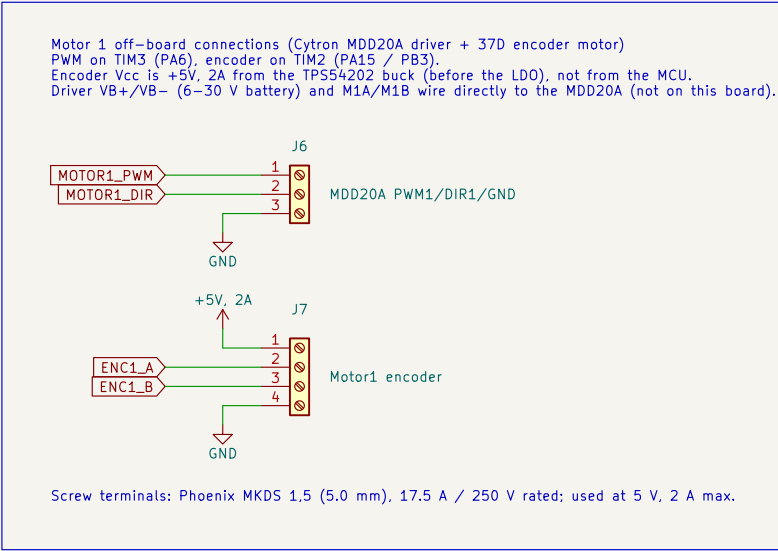
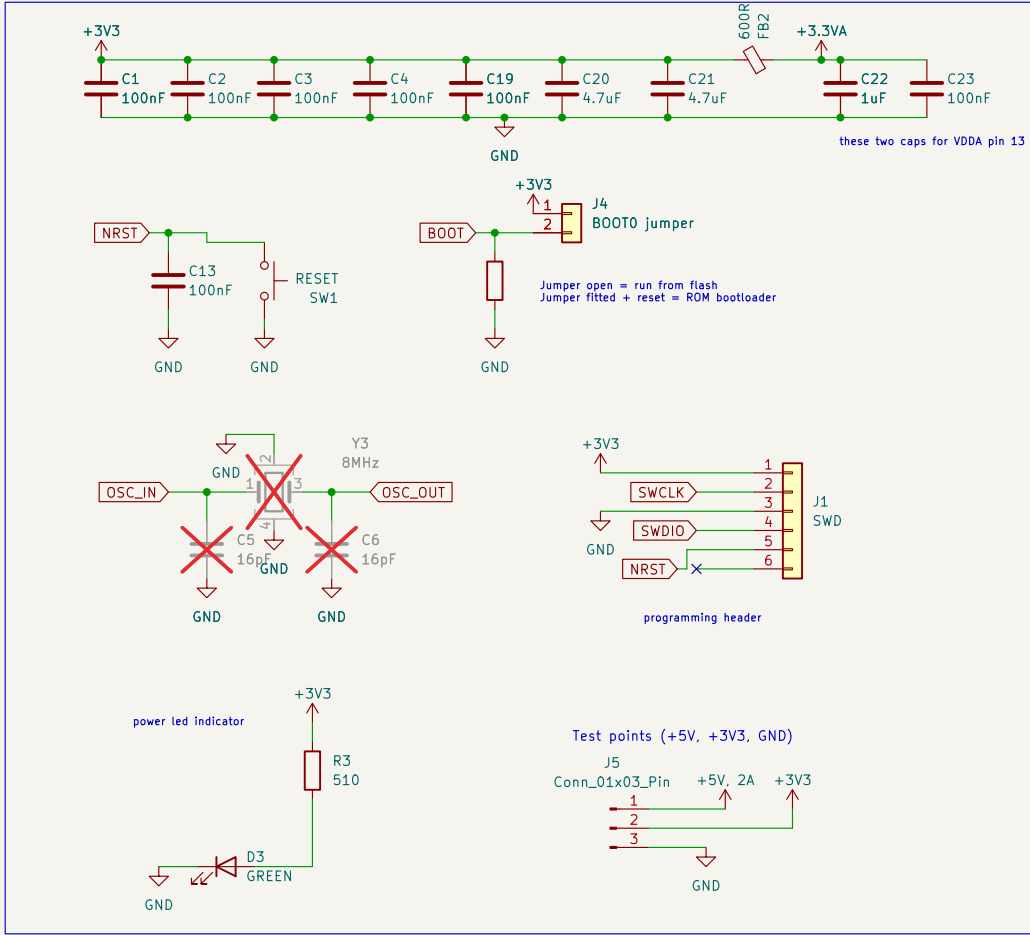
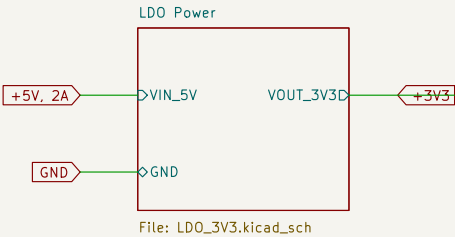
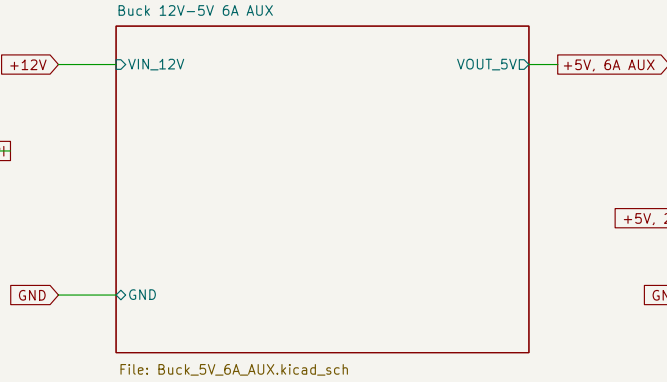
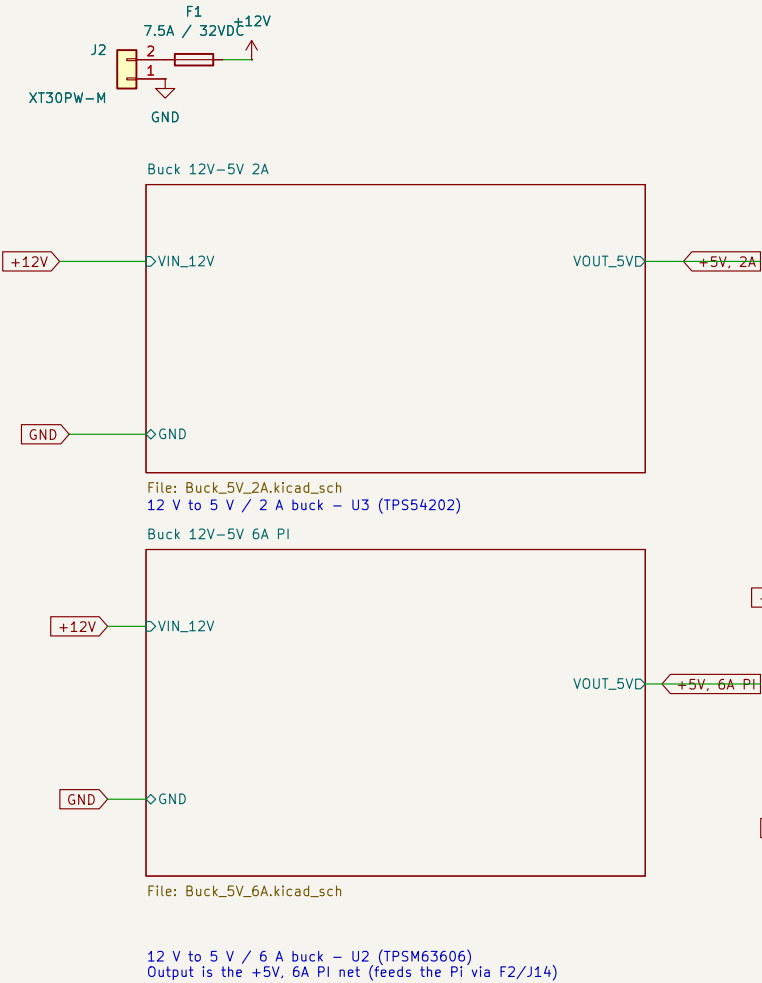
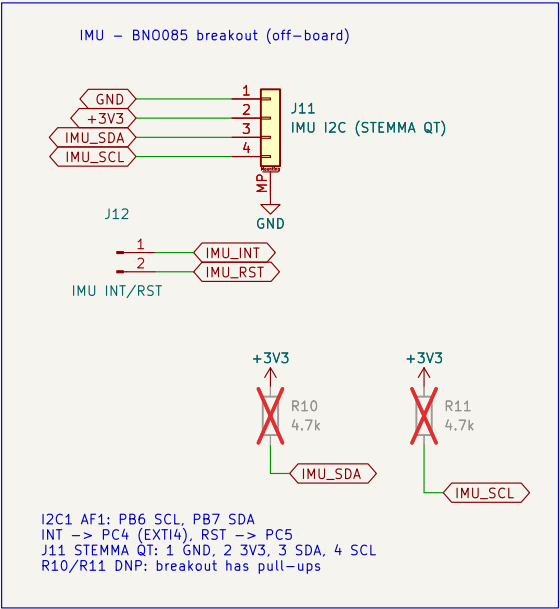
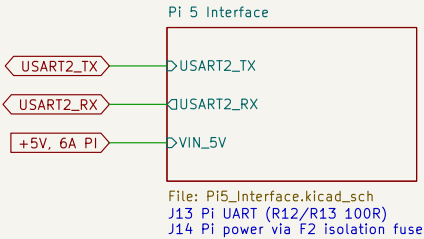
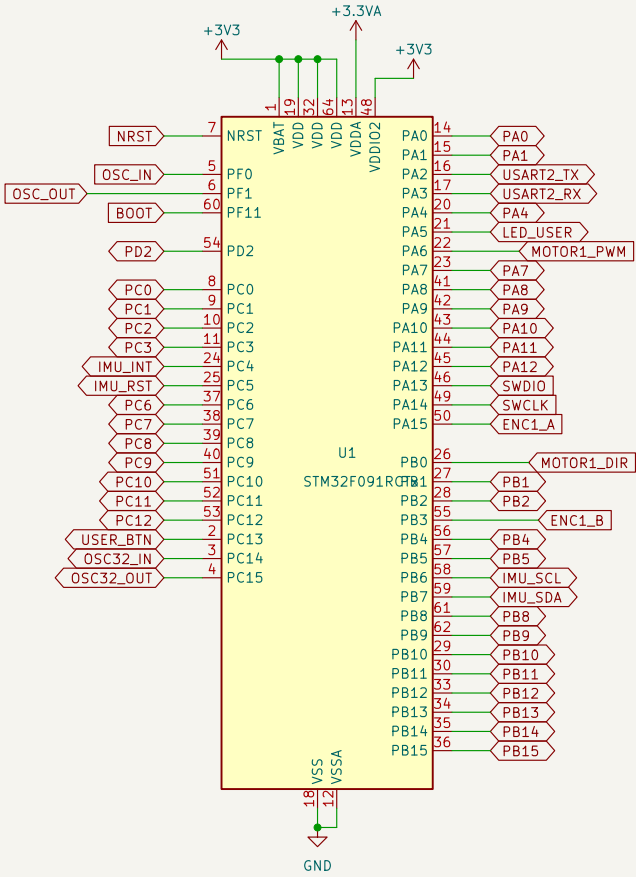




MCU GPIO Breakout

File: GPIO_Breakout.kicad_sch
J8-J10: 2.54 mm headers for every spare STM32 pin.
Connected by global labels, so the block has no sheet pins.

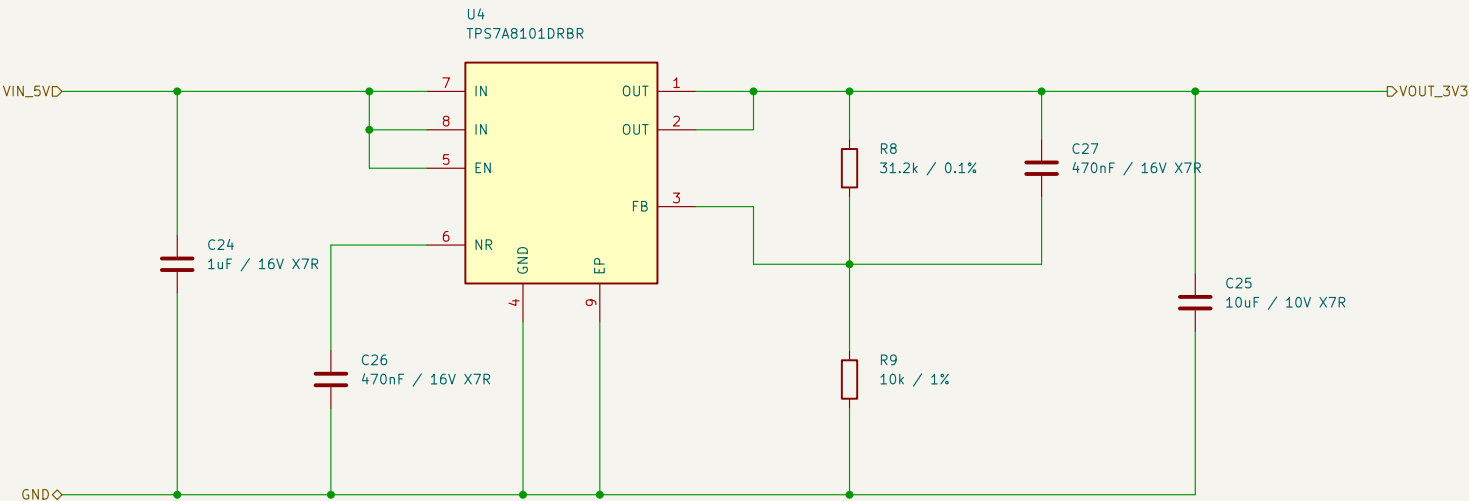


Delivery robot senior design - MCU core
Derived from ST NUCLEO-64 MB1136 rev C-04, sheets 2-3

Sheet: / File: delivery_robot_1.kicad_sch		
Title: STM32F091RC core - ready for peripherals		
Size: A3	Date: 2026-09-24	Rev: 1.0
KiCad E.D.A. 10.0.6		Id: 1/7

5 V to 3.3 V LDO supply

Reusable sheet: VIN_5V input, VOUT_3V3 output, common GND



VOUT nominal = $0.8 \times (1 + 31.2k / 10k) = 3.296 \text{ V}$
1 A is the TOTAL output budget for STM32 + IMU + other 3.3 V loads.

C2: require $\geq 4.7\mu\text{F}$ EFFECTIVE capacitance; selected part modeled at 8.8uF.
EN tied to IN. C3 sets noise reduction / soft start; C4 bypasses R1.

Layout: capacitors close to U1; EP pad 9 to GND copper and thermal vias.
Heat at 1 A: about 1.7 W at 5 V input; verify junction temperature on final PCB.

5 V motor / encoder supply branches BEFORE this sheet.
Adafruit BNO085 breakout: 3.3 V rail to VIN; 3Vo is an output, leave separate.
Bare BNO085: connect 3.3 V to VDD and VDDIO with its specified decoupling.

5 V to 3.3 V, 1 A total rail capacity

Senior Design

Sheet: /LDO Power/
File: LDO_3V3.kicad_sch

Title: LDO power supply – reusable sheet

Size: A4

Date: 2026–10–05

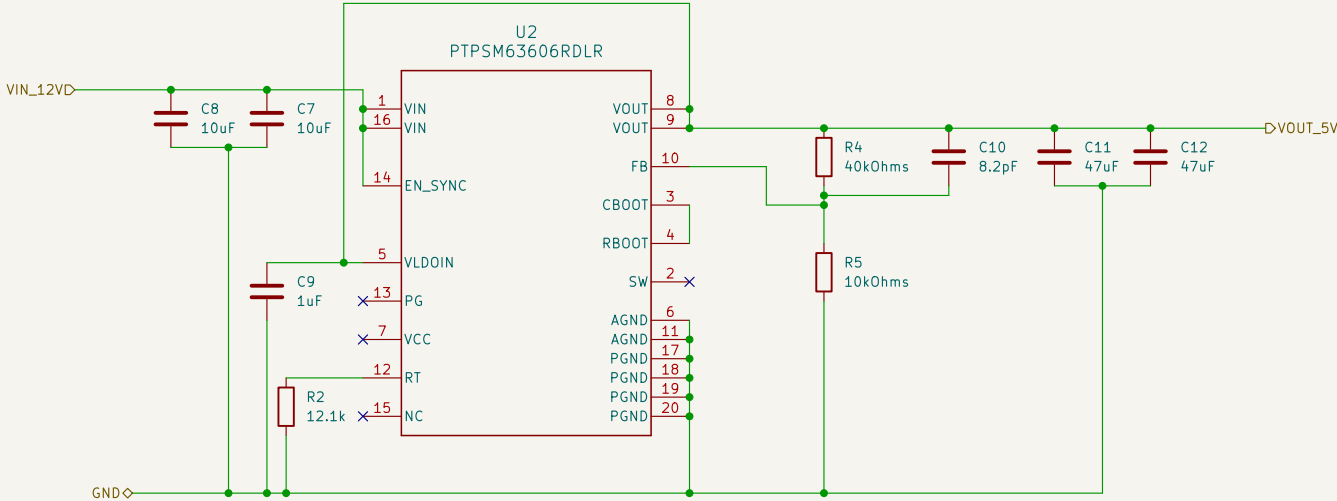
Rev: 1.0

KiCad E.D.A. 10.0.6

Id: 2/7

12 V – 5 V, 6A Buck Convertor

12 V to 5 V buck converter – TPSM63606 (U2)



Reusable sheet: VIN_12V input, VOUT_5V output, common GND
VOUT set by R4/R5 divider on FB; RT = R2; EN tied to VIN.
RBOOT shorted to CBOOT; SW left unconnected (inductor is inside the module).

12 V to 5 V buck, TPSM63606 (U2), 6 A rail

Senior Design

Sheet: /Buck 12V-5V 6A PI/

File: Buck_5V_6A.kicad_sch

Title: Buck converter – reusable sheet

Size: A4

Date: 2026-10-06

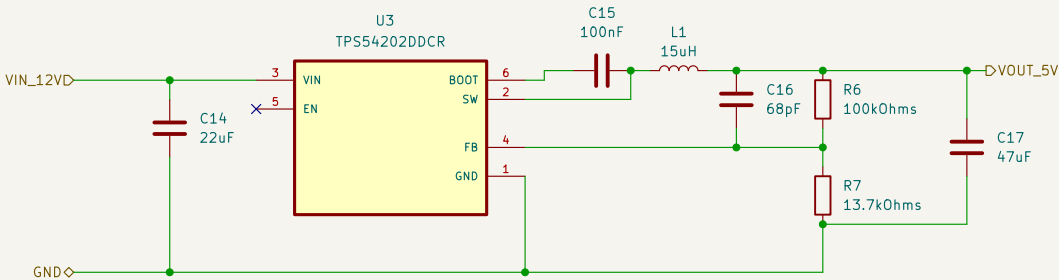
Rev: 1.0

KiCad E.D.A. 10.0.6

Id: 3/7

12 V – 5 V, 2A Buck Convertor

12 V to 5 V buck converter – TPS54202 (U3)



Reusable sheet: VIN_12V input, VOUT_5V output, common GND

12 V to 5 V buck, TPS54202 (U3), 2 A rail

Senior Design

Sheet: /Buck 12V–5V 2A/

File: Buck_5V_2A.kicad_sch

Title: Buck converter – reusable sheet

Size: A4

Date: 2026–10–06

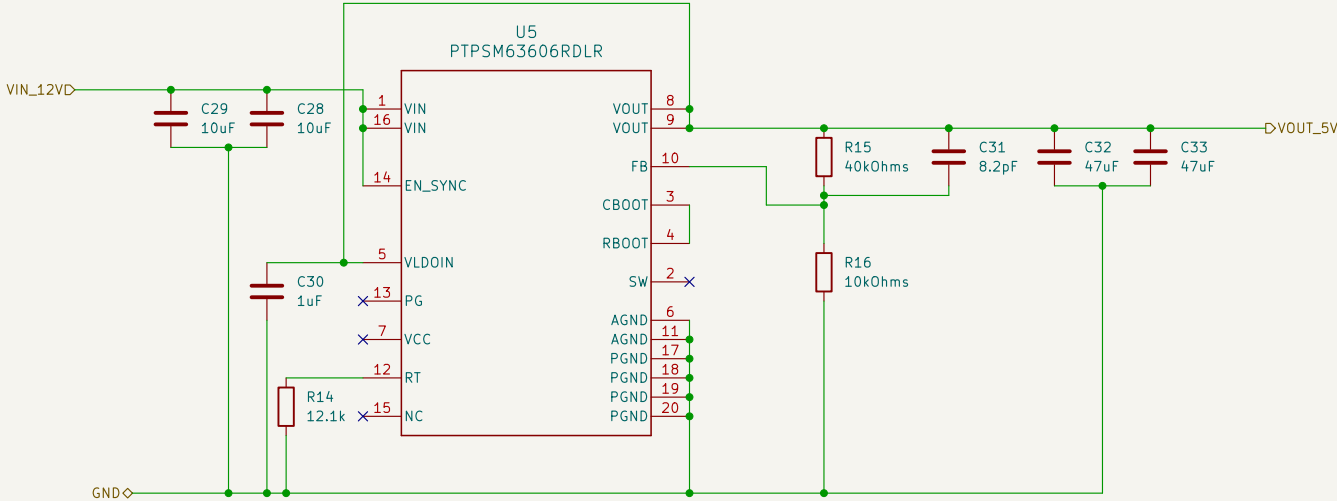
Rev: 1.0

KiCad E.D.A. 10.0.6

Id: 4/7

12 V – 5 V, 6A Buck Convertor

12 V to 5 V buck converter – TPSM63606 (U5)



Reusable sheet: VIN_12V input, VOUT_5V output, common GND.
VOUT set by R15/R16 divider on FB; RT = R14; EN tied to VIN.
RBOOT shorted to CBOOT; SW left unconnected (inductor is inside the module).

12 V to 5 V buck, TPSM63606 (U5), 6 A rail

Senior Design

Sheet: /Buck 12V-5V 6A AUX/

File: Buck_5V_6A_AUX.kicad_sch

Title: Buck converter – reusable sheet

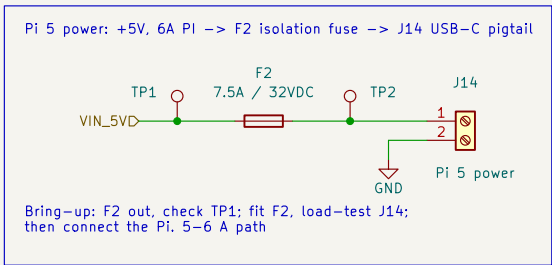
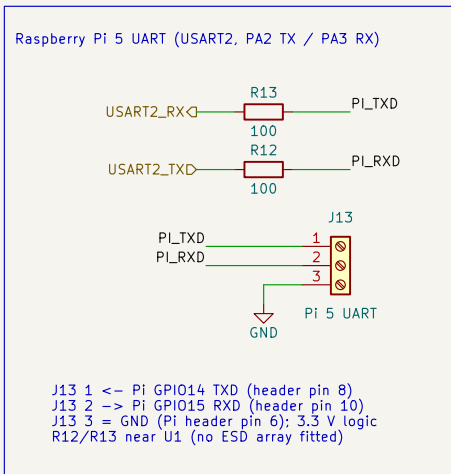
Size: A4

Date: 2026-10-06

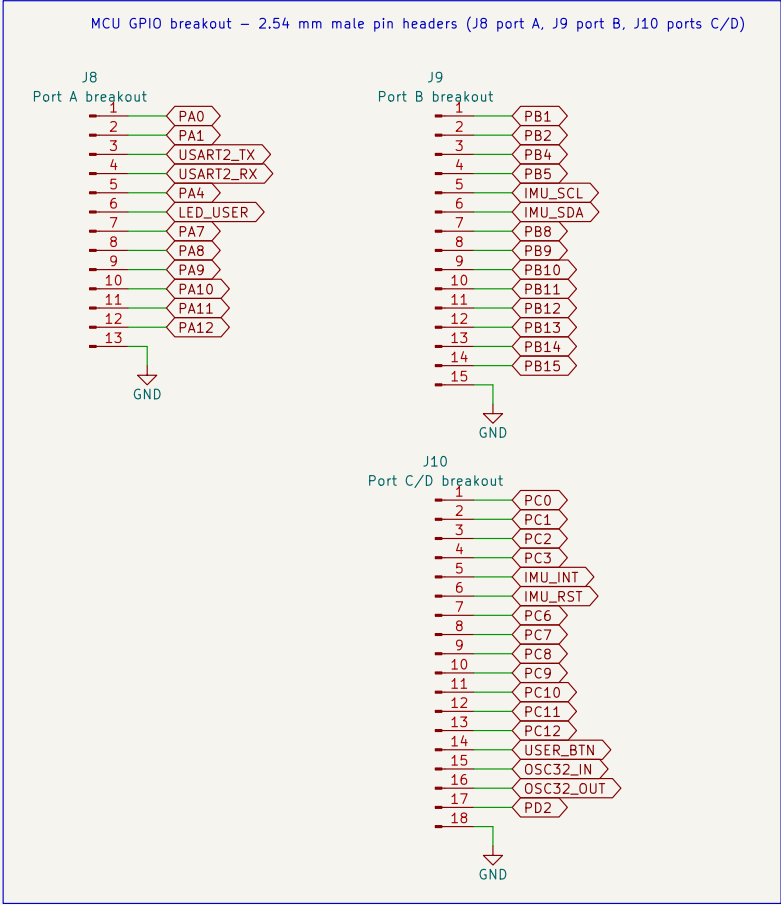
Rev: 1.0

KiCad E.D.A. 10.0.6

Id: 5/7



Senior Design		
Sheet: /Pi 5 Interface/		
File: Pi5_Interface.kicad_sch		
Title: Raspberry Pi 5 interface – UART and power		
Size: A4	Date: 2026-10-07	Rev: 1.0
KiCad E.D.A. 10.0.6		Id: 6/7



Senior Design

Sheet: /MCU GPIO Breakout/
File: GPIO_Breakout.kicad_sch

Title: STM32 GPIO breakout headers

Size: A4	Date: 2026-10-07	Rev: 1.0
KiCad E.D.A. 10.0.6		Id: 7/7